

# "I Am a Blind Seller!": Picture Taking Assistance for Visually Impaired Individuals for Participation as Sellers in Customer to Customer (C2C) Marketplaces

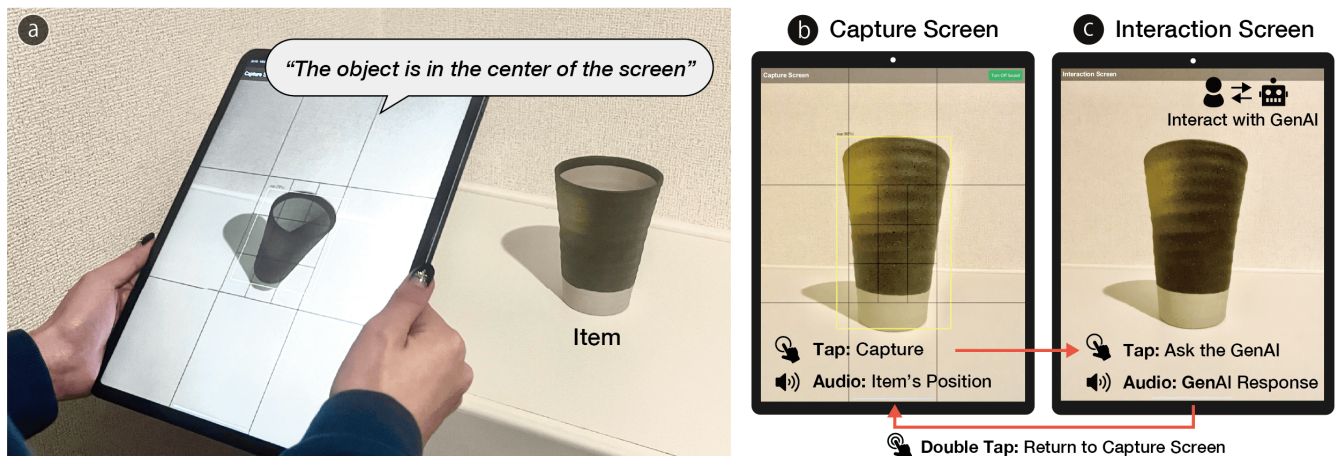
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**Figure 1:** We introduce our prototype designed to enable BLV individuals to participate as sellers on C2C marketplaces. The prototype offers composition assistance during photography and interactions using GenAI. (a) Taking a photo of a product (a cup) for sale. The screen is divided into a grid of nine rectangles, with the central rectangle subdivided further into a smaller grid. Once an object is recognized, a bounding box appears and the position of its center is provided via audio feedback. (b) On the capture screen users locate an item using audio feedback and tap to capture the photo once it is centered. (c) On the interaction screen, users can ask GenAI questions about the captured item, such as color and shape, using voice commands.

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## Abstract

Consumer-to-consumer (C2C) marketplaces have grown as platforms for buying and selling products, driven by interest in sustainable and cost-effective shopping. However, the reliance on visual communication, especially product photos, presents barriers for visually impaired (BLV) individuals who often struggle to capture high-quality images independently. To identify the challenges of product photography, we conducted semi-structured interviews with  $N = 6$  BLV participants. Based on the findings, we developed a prototype that integrates real-time object detection for audio-assisted composition during photography and interaction with

Generative AI (GenAI) after capture. We evaluated the prototype with  $N = 4$  BLV participants, comparing it to conventional photography applications. Our findings highlight the unique demands of photography in C2C marketplace and demonstrate the potential of GenAI tools to empower BLV sellers to participate effectively and independently.

## CCS Concepts

• **Human-centered computing** → **Accessibility design and evaluation methods.**

## Keywords

Photo Assistance, Accessibility, Generative AI, Design, Blind or low vision

### ACM Reference Format:

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## 1 INTRODUCTION

Consumer-to-consumer (C2C) marketplaces, such as eBay<sup>1</sup>, Mercari<sup>2</sup> and Depop<sup>3</sup>, offer accessible platforms for buying and selling products and have gained increasing popularity amid growing interest in sustainable and cost-effective shopping. However, these platforms are highly dependent on visual communication to convey the appeal of products, posing significant participation barriers for blind and low vision (BLV) users. Despite the desire of BLV users to participate in C2C marketplaces [28], they often struggle to receive adequate support for taking product photos or accurately adjusting compositions due to limited visual feedback [34].

Numerous assistive technologies have been developed to address the challenges BLV users face during daily photography. These include systems providing real-time audio guidance for framing using regions of interest [38], composition support algorithms based on audio and vibration feedback [19], and methods offering real-time feedback on subject position and size [4]. However, these technologies often restrict subjects to humans or are limited to everyday scenarios, highlighting the need for more versatile support across diverse contexts. Our study specifically addresses the underexplored context of sellers on C2C platforms, where BLV individuals can enter at low cost and engage in social participation. On these platforms, conveying the visual appeal of products is crucial, making it essential to position items centrally and clearly [35]. However, for BLV individuals, effectively framing products and obtaining the necessary information remains a significant hurdle.

Common Artificial Intelligence (AI)-powered tools such as Be My Eyes [12], TapTapSee [36], and Seeing AI [26] offer real-time assistance and object recognition for BLV users. While they are effective in providing general feedback, these applications often deliver information without prioritizing or filtering details based

on the user's immediate needs. This lack of customization means users receive all available information about an object, which can include unnecessary details that confuse rather than clarify. Since these tools also do not offer interactive dialogue, users cannot easily refine or specify their inquiries, leading to an overwhelming influx of information and making it difficult to extract the precise details needed. Furthermore, the absence of composition assistance means the tools fail to guide objects that are out of frame, necessitating the switch between multiple apps and resulting in an inconsistent user experience.

Given these challenges, we propose adopting a GenAI solution, as it allows for multimodal input and can provide real-time, context-aware feedback to user inquiries, facilitating multimodal interactions [3]. This enables users to ask specific details about products, such as color and orientation, and obtain accurate information tailored to their needs. Furthermore, by utilizing GenAI, users can experience a seamless process from composition assistance to information gathering on a single platform without switching between tools. This empowers BLV users to have more effective, independent product sales on C2C platforms, offering customization and interactive feedback not achievable with traditional AI tools, as well as providing a solution that fundamentally addresses the challenges faced by BLV individuals.

To address this gap, our study focuses on two key Research Questions (RQ):

**RQ1:** What challenges do BLV users face when photographing items for sale on C2C marketplaces?

**RQ2:** How does the proposed prototype compare to existing tools in supporting BLV sellers during product photography?

To answer these questions, we first conducted semi-structured interviews with  $N = 6$  BLV participants to identify specific challenges they face when photographing items for C2C platforms (RQ1). Based on the insights gained, we developed a prototype that provides composition support using object detection during photography and interactive assistance with GenAI after photo capture. While taking a picture, users receive audio feedback for real-time alignment, and afterwards, GenAI enables interactive engagement with the images, allowing users to ask specific questions and receive tailored audio feedback. We then evaluated the effectiveness of our prototype by comparing it with a default iOS camera application running on iPad with VoiceOver enabled, Seeing AI application, and our prototype in a scenario where  $N = 4$  BLV participants were tasked with taking photos as sellers (RQ2). The results showed that our prototype enables users to effectively frame items and gather information tailored to their needs, enhancing their confidence in photography for sales purposes. By addressing these challenges, our research aims to enhance accessibility to C2C platforms and enable BLV individuals to fully participate as sellers.

## 2 RELATED WORK

### 2.1 Photography Assistance Technologies for BLV Individuals

BLV individuals often take photographs to connect and communicate with close friends or post on social media [6]. However, when taking a picture they oftentimes face difficulties with subject

<sup>1</sup><https://www.ebay.com/>

<sup>2</sup><https://www.mercari.com/>

<sup>3</sup><https://www.depop.com/>

visibility and framing [1, 13, 16]. Despite these challenges, BLV individuals desire to take photos independently, without relying on others [6]. Consequently, various research efforts have been undertaken to simplify the photography process for BLV individuals. For instance, VisPhoto captures a 360° view and allows users to crop the desired subject [18]. While this removes the need to search for the subject, the delay between capture and processing requires users to remember the target and scenario of the photo. Applications providing audio and vibration feedback on camera tilt and subject size have also been developed to assist in photography [19, 33, 38, 39, 41]. However, these tools are often limited to information such as the number of faces or face sizes within a photo, and the shooting scenarios are mostly restricted to everyday life, indicating a need to explore support across diverse contexts.

Existing solutions for composition focus on central and golden ratios [5] and are primarily purposed for human portrait photography, addressing consistency issues with face detection algorithms. As for confirming captured subjects, there are approaches that integrate ambient audio and alerts to aid with visual recall of photos [16]. However, since ambient noise is also recorded, the cues for recalling context can become complicated. These studies collectively reveal a gap where delays in capturing and processing increase cognitive load for users, highlighting the necessity for real-time capabilities that allow BLV users to accurately operate cameras and access item information. Existing photography support is largely confined to everyday scenarios and lacks specialized applications, such as picture taking for listing items for sale online. Thus, our research aims to empower BLV individuals in C2C marketplaces by supporting effective product photography and sales, expanding their opportunities for economic independence and social participation.

## 2.2 Advancements in GenAI and Interactive Feedback Systems

AI technologies designed to assist BLV individuals have gained attention in recent years. For example, Seeing AI [26], Envision AI [11], TapTapSee [36] and Lookout [24] provides real-time object recognition and environmental descriptions. Similarly, Be My Eyes supplements visual information through collaboration with AI and volunteers [12]. While these tools efficiently provide general visual assistance, they do not directly support the picture taking process. As a result, users must switch between multiple tools, using one for creating listed product description and another for taking a product photo, increasing overall cognitive workload and complexity of the process. However, advancements in GenAI have opened new possibilities for visual tasks. Models like DALL-E [7] can generate images from text, and GPT (Generative Pretrained Transformer) based models enable sophisticated natural language interactions [21]. These technologies hold a great potential for real-time photo-taking assistance and interactive feedback, especially in the context of C2C marketplace product photography, a field that remains relatively unexplored.

Furthermore, computer vision models, such as YOLO (You Only Look Once) and SSD (Single-Shot Detector), have demonstrated capabilities in real-time object detection, accurately identifying position and features [23, 32]. These technologies lay a foundation for

composition adjustments and real-time feedback. However, their integration into C2C marketplaces to support the BLV individuals is still a relatively unexplored area of research. In addition, there are advancements in using GenAI for real-time visual content descriptions, allowing users to obtain image descriptions on the fly [14, 17, 27, 30, 40, 42]. Systems that generate customizable visual descriptions based on user context [10] and technologies that customize audio descriptions to enhance viewing experiences for BLV individuals [29] also show promise. Integrating these technologies into C2C marketplaces could open new avenues for BLV individuals to effectively sell products.

## 3 USER STUDY 1: FORMATIVE STUDY

To clearly identify the challenges faced by BLV users when selling on C2C marketplaces and directly address the RQ1, we conducted a series of semi-structured interviews with N = 6 BLV participants.

### 3.1 Participants and Method

Using snowball sampling, we recruited N = 6 BLV participants (all male, mean age = 32.83, SD = 12.01), of whom three were completely blind and three had low vision impairment. Two participants (P1, P3) had experience as sellers on C2C platforms, but received assistance from family members. Other four participants (P2, P4, P5, and P6) attempted to sell, but stopped due to a lack of assistance in picture taking process. Detailed participant information is provided in Table 1. Each participant took part in a 60-minute interview conducted via Zoom<sup>4</sup>, which was recorded for data analysis. Only participants who consented to the experiment were included, and they received a \$30 reward. This study was approved by the Institutional Review Board (IRB).

The interviews began with background questions regarding the participants' selling experiences on C2C marketplace, followed by questions about the challenges they faced when photographing items for listing. The recorded scripts were transcribed using Notta<sup>5</sup>. We conducted a reflexive thematic analysis [8] across six phases. Two authors analyzed the data. Coding was performed using digital sticky notes in Miro<sup>6</sup>. For example, a participant's statement, "I take photos by intuition," was treated as a single code. Similar sentiments from other participants were tagged with the same code. This code and related codes were grouped under the theme "The Difficulty of Correctly Placing Items in the Frame". We identified 42 codes. These were then categorized using the affinity diagram technique [25] into three categories under the main theme of "Challenges in Photographing Items for Listings".

### 3.2 Findings: Challenges in Photographing Items for Listings

All participants encountered challenges when taking photos for sales. These difficulties included not being able to fit items within the frame, needing to switch between multiple applications, and lacking information about the products. To overcome these obstacles, participants relied on family members, followed their intuition,

<sup>4</sup><https://zoom.us/>

<sup>5</sup><https://www.notta.ai/en>

<sup>6</sup><https://miro.com/ja/login/>

or utilized AI tools like Seeing AI [26], each adopting their own unique strategies.

**The Difficulty of Correctly Placing Items in the Frame.** N = 3 participants noted that they often rely on intuition when taking photos, which frequently results in items being out of the frame or placed inappropriately. P2 emphasized, *“I take photos by intuition, but they always end up outside the frame,”* highlighting the need for compositional assistance during photography. Additionally, P6 stated, *“Photos for sale need to be neatly centered, but conventional apps only provide rough positioning guidance. Without detailed compositional assistance, it’s hard to get the item properly framed.”* This indicates that detailed composition assistance is necessary for product photography intended for sales.

**Cross-Application Usage.** N = 3 participants highlighted that using multiple apps to take a single sales photo requires considerable effort. P6 stated, *“There are apps that guide composition, but they don’t allow photo capture simultaneously. While there are features for AI to describe the photos, there’s no support for taking the photo in the first place,”* indicating the need for simultaneous support in item recognition and composition. Similarly, P4 mentioned, *“First, I enlarge the item, then take a photo and ask AI for an explanation. It’s cumbersome to use multiple tools just to take a photo for selling.”* P3 also commented, *“With the iPhone camera using VoiceOver, people are detected but objects aren’t, so I need to find a different method for product photography.”* These responses reveal the necessity of integrating composition assistance, photo capture, and item information presentation into a single application.

**Verification of Captured Items.** N = 3 participants (P1, P3, and P5), who are low vision users, noted their ability to generally see the items and aim for central composition with their residual vision. However, they could not perform detailed checks of the photos. P1 and P5 shared, *“I can see overall shapes and position items centrally, but I cannot distinguish the front from the back of the product.”* P3 added, *“I cannot tell if there is glare in the photos.”* Both P1 and P3 mentioned relying on family to review the photos before listing the items for sale. Although they can independently take photos, they require assistance to ensure the detailed information and quality needed for successful selling, such as verifying the condition of the item, ensuring accurate color representation, and confirming text clarity on packaging.

## 4 PROTOTYPE DESIGN

Our prototype integrates item recognition and composition support into a single application, allowing BLV users to capture images and receive detailed product explanations. It includes GenAI-powered features for summarizing visual information and facilitating Q&A interactions, enabling users to question their images and receive immediate responses, thus streamlining the entire process from photography to information gathering for autonomous selling activities.

### 4.1 System Architecture and Functionality

The prototype, illustrated in Figure 2, provides real-time object detection and GenAI-driven conversational features, utilizing Next.js as its back end. It consists of two main pages: the capture screen

and the interaction screen (Figure 1 (b), (c)). The capture screen uses the device’s camera to acquire a video stream for real-time object detection, rendered on a canvas. The COCO-SSD model [23] detects objects, enabling BLV users to intuitively grasp object positions. The interaction screen allows users to query GenAI based on captured images, receiving prompt and detailed descriptions via OpenAI’s GPT-4o mini [31].

### 4.2 Object Detection Pipeline

The capture screen performs object detection with the COCO-SSD library, updating every 100 milliseconds [22]. The camera screen is divided into a 3 x 3 grid, with the center further divided into a 3 x 3 grid. Recognized objects are displayed with bounding boxes, and their center positions are provided via audio feedback using the Web Speech API. BLV users can capture frames by tapping, storing images for later GenAI interaction, with a shutter sound indicating successful capture.

### 4.3 AI Interaction Pipeline

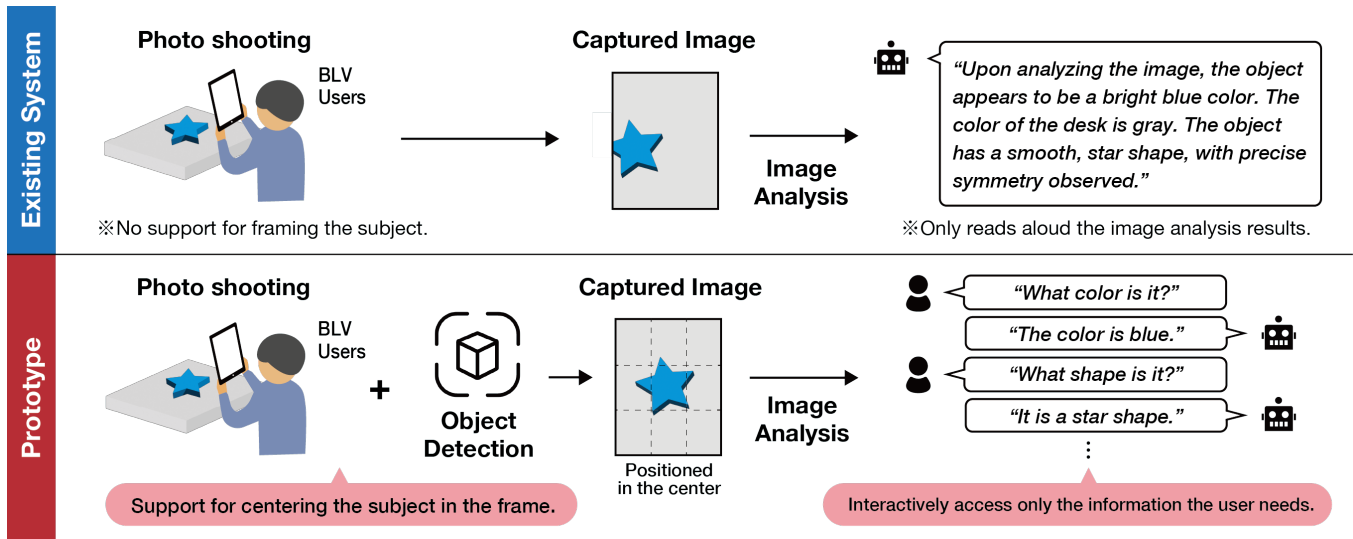
BLV users engage with GenAI on the interaction screen, where voice recognition converts speech to text, enabling them to use the interface without relying on sight. Upon tapping, a sound confirms activation, and the queried content is sent to the backend via OpenAI’s API. Once the analysis is complete, responses are read aloud. Users can also return to the capture screen with a double tap. This prototype facilitates interactive GenAI interaction, in contrast to previous tools that provide one-way image information.

## 5 USER STUDY 2: EVALUATION OF THE PROTOTYPE

The user study aimed to evaluate the effectiveness of our GenAI-powered prototype in helping BLV individuals photograph products for C2C marketplaces, directly addressing RQ2. By comparing our prototype with the default iOS camera application running on iPad and the Seeing AI application [26], which is commonly used by BLV individuals, we assessed its ability to assist in capturing well-aligned and detailed images. We used eye drop bottles as subjects because they lack tactile cues, presenting a challenge that requires advanced assistance.

### 5.1 Participants and Method

We conducted a one-hour in-person study with N = 4 BLV users (2 male and 2 female, with a mean age = 21.5, SD = 1.12) presented in Table 2). Of the participants, one was blind and three had low vision, with varying visual fields. All participants regularly used smartphones for everyday photography and had experience with the VoiceOver feature and Seeing AI. Since none of the participants had taken part in User Study 1 (Section 3), they were first briefed on the challenges of product photography in sales contexts. Then, we explained the purpose of the experiment to the participants and allowed them to practice taking photos with three tools: the default camera application on an iPad with VoiceOver (12.9 inch 4th generation) running iOS (version 17.5.1), the Seeing AI application, and our prototype. Participants were then asked to photograph eye drops under the scenario, *“You are attempting to sell eye drops. As a seller, you need to position the product centrally and clearly.”* Each



**Figure 2:** The top section illustrates an example of existing systems, while the bottom section shows the outline of the prototype. The existing system provides detailed descriptions of the captured items but lacks features to assist with composition during photography. Additionally, it cannot tailor the explanation content to the user's needs. Our prototype, on the other hand, allows for composition support and GenAI-powered questioning all within a single tool.

participant captured three photos, using the iPad camera, Seeing AI, and the prototype in that order. Although our prototype allows interaction with GenAI, there was no specified limit to the number of interactions. After photographing, participants completed the System Usability Scale (SUS) [9] evaluation for each tool, as well as a 5-point Likert scale assessment for six questions shown in Table 3. Finally, we conducted interviews to gather feedback on overall preferences regarding each tool. This experiment was conducted with the approval of the IRB and was only carried out with participants who gave their consent.

## 5.2 Result

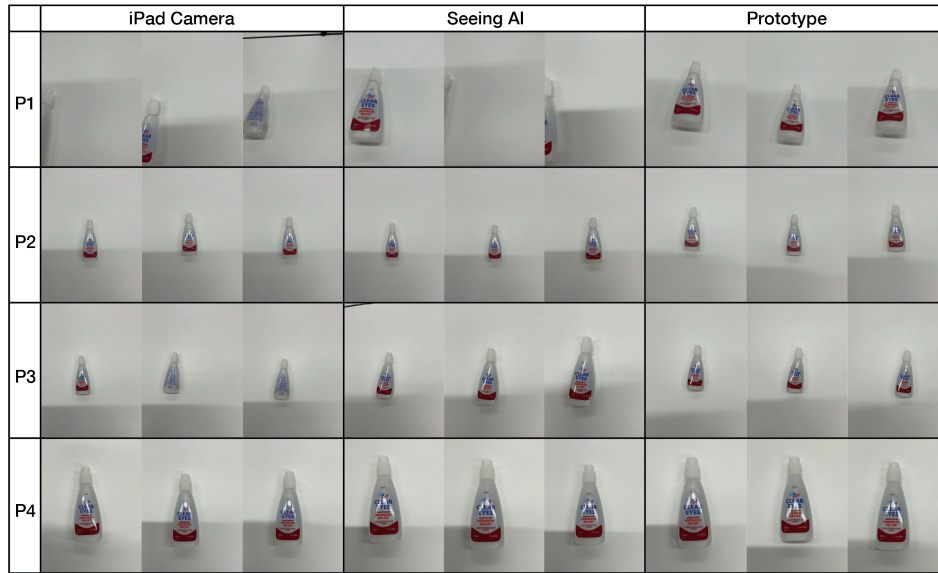
Participants quickly mastered the functions of the developed prototype and were able to take photos with ease. Figure 3 shows the photos of the eye drop taken by participants using the three tools. Participants with residual vision (P2, P3, P4) were able to center the items and take photos regardless of the tool used. In contrast, the participant with no residual vision (P1) struggled to center the item using the iOS camera application running on iPad and Seeing AI application. Additionally, both P1 and P3 were unable to determine the front and back of the eye drop bottle when using the unassisted iPad camera, preventing them from orienting the item correctly for the photograph.

The average total time taken to capture three photos was approximately 46 seconds with the iPad, 227.25 seconds with Seeing AI, and 312.5 seconds with the prototype. Seeing AI has a feature that reads out details of the captured photo, and the prototype allows users to ask questions about the photo. Therefore, the amount of information read aloud and the number of questions asked can affect the duration of use. The longer average time with the prototype indicates that there is significant interaction between participants and the AI during the photo-taking process.

**5.2.1 Measuring Deviation from The Frame Center.** We evaluated the accuracy of the deviation from the central position of objects in images captured using three different methods (Figure 3). The images were standardized to a size of 720 x 940 pixels, and the object's position was determined by the center of the bounding box obtained through object recognition. The offset distance from the image center to the object center was calculated in pixels. If the object could not be recognized, the offset was set to 360 pixels (half of the 720-pixel frame width). The average offset calculated using this method was 127.49 pixels for the iPad camera, 122.99 pixels for Seeing AI, and 46.49 pixels for the prototype. These results indicate that the prototype has a higher capability to capture objects closer to the center than the standard camera and Seeing AI.

**5.2.2 SUS Results for Each Tools.** The average overall SUS score was 56.25 for the iPad camera, 70 for Seeing AI, and 73.12 for the prototype. According to the SUS benchmarks [9], the Adjective Rating for the iPad default camera application was classified as "Poor," while both Seeing AI and the prototype were rated as "Good." The SUS results highlighted that the interaction facilitated by the prototype between humans and GenAI enhances accessibility, enabling participants to easily gather necessary information about the product. Furthermore, participants suggested that the prototype functions not only as a photography tool but also as an information-gathering tool, confirming its effectiveness in unique sales situations.

**5.2.3 5-point Likert Scale Evaluation.** In the 5-point Likert scale evaluation, participants rated the prototype as the most user-friendly tool for sales, followed by Seeing AI, and finally the iPad camera, as shown in Figure 4. The iPad camera application scored the lowest due to its lack of object recognition assistance and informational support, which made the interface feel inconsiderate of BLV users'



**Figure 3: Participants took photos of eye drops using the iPad camera, Seeing AI, and the prototype. Each participant repeated this process three times.**

needs. Participants found it stressful and unenjoyable to use, especially when trying to identify features such as orientation and color. This inability to capture appealing product photos for potential buyers led to difficulties in using the iPad camera for item listings. In contrast, Seeing AI received praise for its thoughtful design for BLV users, but presented issues with information overload and lack of composition assistance, causing uncertainty among some participants about its effectiveness for sales photography. The prototype was rated the highest, recognized for its intuitive interface and efficient composition assistance, which improved participants' confidence in framing subjects and verifying product details effectively.

**5.2.4 Interview Findings.** The interviews revealed that participants found the prototype to be the most user-friendly tool, mirroring the Likert scale findings. They expressed frustration with existing tools like Seeing AI, which often provided excessive information that led to irritation and forgetfulness of critical details. The prototype, on the other hand, was appreciated for its succinct and direct responses, allowing users to manage inquiries and request details as needed. P4 found it convenient to understand item features and obtain translations, highlighting the value of intuitive human-computer interaction facilitated by GenAI. Moreover, the integration of photography and GenAI reduced the need to switch between applications and eliminated unnecessary cropping through composition assistance. P1's comments on the importance of having assistance before obtaining detailed explanations underscored this necessity. Ultimately, the prototype expanded participants' ability to sell independently, increasing their confidence and illustrating the significant role of composition aid in creating compelling sales photos. Participants expressed newfound independence, as P1 remarked, *"I used to have my parents check the photos I took, but now I feel like I can do it by myself."*

## 6 DISCUSSION AND LIMITATION

This study used non-embossed eye drops as the sole test item, but real-world sale items vary [43]. While GenAI effectively identified item fronts and backs beyond tactile feedback, further evaluation with a variety of item sizes, shapes, and textures is needed to enhance prototype versatility. The study did not involve actual selling, necessitating more practical evaluations. Additionally, determining if prototype-taken photos attract buyers requires interface updates. Successful product photos may depend on lighting, color, and shadow adjustments, making it essential to identify effective image characteristics on C2C platforms. Our use of the COCO-SSD model supports 80 object types [37], but further research should compare other models to find the most suitable for sales. Given the limited participants, cautious generalization is advised, and including a diverse, larger participant pool can enhance statistical reliability.

In parallel, addressing the inherent randomness and unpredictability of GenAI [20] is crucial, as it risks presenting misleading contexts. Studies show that BLV users may struggle to fully trust GenAI-derived results [2], emphasizing the need for careful task allocation between humans and GenAI, especially in sales. Privacy and security require improved management of information captured in image backgrounds [15]; the current focus on central composition support needs careful consideration in real-world applications.

## 7 CONCLUSION AND FUTURE WORK

We developed a prototype to enable BLV individuals to become sellers on C2C platforms. This prototype integrates real-time composition assistance via object recognition during photography and employs GenAI for text generation and interactive dialogue post-photography. We conducted semi-structured interviews with six BLV individuals who have experience with or have attempted to sell goods, highlighting the current challenges of photo-taking in

sales contexts. Our prototype allows users to inquire about captured items with GenAI, facilitating enhanced interaction and decision-making. A user study with four BLV participants demonstrated the effectiveness of using the prototype for photography in sales scenarios. Looking ahead, we aim to test the prototype with a more diverse range of products and gather feedback from buyers regarding the photos taken by BLV sellers. Additionally, we plan to identify the characteristics of effective sales photos, incorporate features such as automatic lighting and brightness adjustments, and add the ability to automatically generate product descriptions based on captured images.

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## A APPENDIX

| ID | Gender | Age | Vision Status                       | Experience in selling on C2C marketplaces                |
|----|--------|-----|-------------------------------------|----------------------------------------------------------|
| P1 | Male   | 20  | Cannot see the central part         | Yes (with family support)                                |
| P2 | Male   | 20  | Totally blind (can perceive light)  | Interrupted due to lack of assistance during photography |
| P3 | Male   | 24  | Everything appears white and blurry | Yes (with family support)                                |
| P4 | Male   | 48  | Totally Blind                       | Interrupted due to lack of assistance during photography |
| P5 | Male   | 47  | Visual field of 15 degrees          | Interrupted due to lack of assistance during photography |
| P6 | Male   | 38  | Totally blind, can perceive light   | Interrupted due to lack of assistance during photography |

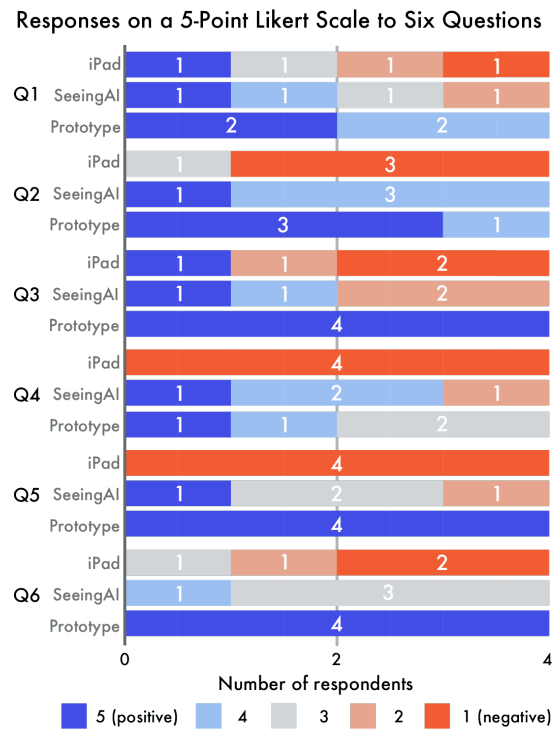
**Table 1: Participant Information for User Study 1.**

| ID | Gender | Age | Vision Status                                        | Commonly used photography tools |
|----|--------|-----|------------------------------------------------------|---------------------------------|
| P1 | Male   | 22  | Blind (only able to perceive light)                  | iPad camera with VoiceOver      |
| P2 | Male   | 23  | Can see overall but everything is blurry             | iPad camera with VoiceOver      |
| P3 | Female | 20  | Can only see the right side of the right eye         | iPad camera with VoiceOver      |
| P4 | Female | 21  | Cannot see the center, but the lower part is visible | Android camera                  |

**Table 2: Participant Information for User Study 2.**

| Number | Content of the Question                                                                                                   |
|--------|---------------------------------------------------------------------------------------------------------------------------|
| Q1     | Was it easy to take photos using this tool?                                                                               |
| Q2     | Did you feel the tool's interface is considerate of BLV users?                                                            |
| Q3     | Did you feel that this tool helps in framing the subjects accurately?                                                     |
| Q4     | Did you feel that this tool appropriately provides detailed information about the subjects (e.g., features and position)? |
| Q5     | Did you enjoy taking photos with this tool?                                                                               |
| Q6     | Would you consider using this tool for selling products?                                                                  |

**Table 3: The content of the six questions asked after taking photos with the iPad camera, Seeing AI, and the prototype in User Study 2.**



**Figure 4: The 5-point Likert scale results of six questions for the three tools: iPad, Seeing AI, and Prototype.**